

LAB FILE -ADVANCED COMPUTER NETWORKS

PCS223

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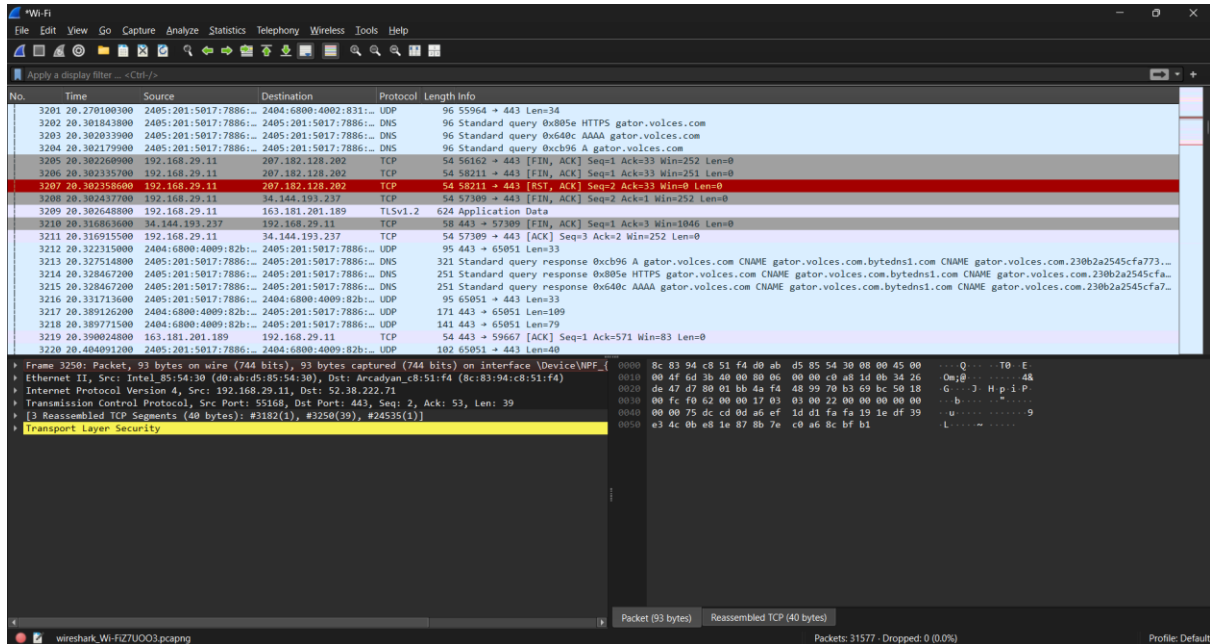
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ASSIGNMENT 1

To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.

Start Packet Capture:



Generate Network Traffic

```
Microsoft Windows [Version 10.0.26200.8246]
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C:\Users\Aryan Kuhar>ping google.com

Pinging google.com [2404:6800:4002:830::200e] with 32 bytes of data:
Reply from 2404:6800:4002:830::200e: time=15ms
Reply from 2404:6800:4002:830::200e: time=14ms
Reply from 2404:6800:4002:830::200e: time=15ms
Reply from 2404:6800:4002:830::200e: time=15ms

Ping statistics for 2404:6800:4002:830::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 15ms, Average = 14ms

C:\Users\Aryan Kuhar>
```

Apply Basic Filters

Use the **Display Filter** bar:

Filter	Purpose
ip	Show IP packets
tcp	Show TCP packets
udp	Show UDP packets
dns	Show DNS packets
icmp	Show ping packets

ip						
No.	Time	Source	Destination	Protocol	Length Info	
3234	20.678348100	163.181.201.189	192.168.29.11	TCP	54	443 → 59667 [ACK] Seq=900 Ack=2682 Win=81 Len=0
3236	20.755832000	172.66.0.227	192.168.29.11	TLSv1.2	105	Application Data
3237	20.762583300	192.168.29.11	172.66.0.227	TLSv1.2	89	Application Data
3238	20.800564200	172.66.0.227	192.168.29.11	TCP	60	443 → 54660 [ACK] Seq=1085 Ack=1402 Win=16 Len=0
3239	20.877449000	163.181.201.189	192.168.29.11	TLSv1.2	1096	Application Data
3240	20.922612600	192.168.29.11	163.181.201.189	TCP	54	59667 → 443 [ACK] Seq=2682 Ack=1942 Win=251 Len=0
3243	22.424136300	192.168.29.1	224.0.0.1	IGMPv3	50	Membership Query, general
3247	22.629495900	192.168.29.11	224.0.0.22	IGMPv3	70	Membership Report / Join group 224.0.0.252 for any sources
3249	23.859291500	52.38.222.71	192.168.29.11	TLSv1.2	106	Application Data
3250	23.859732600	192.168.29.11	52.38.222.71	TLSv1.2	93	Ignored Unknown Record[Illegal Segments]
3407	24.154962100	52.38.222.71	192.168.29.11	TCP	54	443 → 55168 [ACK] Seq=53 Ack=41 Win=199 Len=0
7456	26.111680200	18.66.78.127	192.168.29.11	TLSv1.2	175	Application Data
7457	26.160431500	192.168.29.11	18.66.78.127	TCP	54	64249 → 443 [ACK] Seq=1 Ack=364 Win=1020 Len=0
7458	26.255646400	192.168.29.11	18.97.36.14	TLSv1.2	294	Application Data
7481	26.336935400	172.66.0.227	192.168.29.11	TLSv1.2	100	Application Data
7484	26.345234600	192.168.29.11	172.66.0.227	TLSv1.2	89	Application Data
7485	26.345300300	192.168.29.11	172.66.0.227	TLSv1.2	89	Application Data
7486	26.381454900	172.66.0.227	192.168.29.11	TCP	60	443 → 54660 [ACK] Seq=1131 Ack=1437 Win=16 Len=0
7487	26.381454900	172.66.0.227	192.168.29.11	TCP	60	443 → 54660 [ACK] Seq=1131 Ack=1472 Win=16 Len=0
7488	26.623562300	18.97.36.14	192.168.29.11	TCP	54	443 → 62845 [ACK] Seq=1 Ack=241 Win=12322 Len=0

tcp						
No.	Time	Source	Destination	Protocol	Length Info	
3227	20.610958300	207.182.128.202	192.168.29.11	TCP	54	443 → 58211 [ACK] Seq=33 Ack=2 Win=140 Len=0
3233	20.678128300	163.181.201.189	192.168.29.11	TCP	54	443 → 59667 [ACK] Seq=900 Ack=1254 Win=83 Len=0
3234	20.678348100	163.181.201.189	192.168.29.11	TCP	54	443 → 59667 [ACK] Seq=900 Ack=2682 Win=81 Len=0
3236	20.755832000	172.66.0.227	192.168.29.11	TLSv1.2	105	Application Data
3237	20.762583300	192.168.29.11	172.66.0.227	TLSv1.2	89	Application Data
3238	20.800564200	172.66.0.227	192.168.29.11	TCP	60	443 → 54660 [ACK] Seq=1085 Ack=1402 Win=16 Len=0
3239	20.877449000	163.181.201.189	192.168.29.11	TLSv1.2	1096	Application Data
3240	20.922612600	192.168.29.11	163.181.201.189	TCP	54	59667 → 443 [ACK] Seq=2682 Ack=1942 Win=251 Len=0
3249	23.859291500	52.38.222.71	192.168.29.11	TLSv1.2	106	Application Data
3250	23.859732600	192.168.29.11	52.38.222.71	TLSv1.2	93	Ignored Unknown Record[Illegal Segments]
3407	24.154962100	52.38.222.71	192.168.29.11	TCP	54	443 → 55168 [ACK] Seq=53 Ack=41 Win=199 Len=0
7456	26.111680200	18.66.78.127	192.168.29.11	TLSv1.2	175	Application Data
7457	26.160431500	192.168.29.11	18.66.78.127	TCP	54	64249 → 443 [ACK] Seq=1 Ack=364 Win=1020 Len=0
7458	26.255646400	192.168.29.11	18.97.36.14	TLSv1.2	294	Application Data
7481	26.336935400	172.66.0.227	192.168.29.11	TLSv1.2	100	Application Data
7484	26.345234600	192.168.29.11	172.66.0.227	TLSv1.2	89	Application Data
7485	26.345300300	192.168.29.11	172.66.0.227	TLSv1.2	89	Application Data
7486	26.381454900	172.66.0.227	192.168.29.11	TCP	60	443 → 54660 [ACK] Seq=1131 Ack=1437 Win=16 Len=0
7487	26.381454900	172.66.0.227	192.168.29.11	TCP	60	443 → 54660 [ACK] Seq=1131 Ack=1472 Win=16 Len=0
7488	26.623562300	18.97.36.14	192.168.29.11	TCP	54	443 → 62845 [ACK] Seq=1 Ack=241 Win=12322 Len=0

udp						
No.	Time	Source	Destination	Protocol	Length	Info
3218	20.389771500	2404:6800:4009:82b::...	2405:201:5017:7886::...	UDP	141	443 → 65051 Len=79
3220	20.404091200	2405:201:5017:7886::...	2404:6800:4009:82b::...	UDP	102	65051 → 443 Len=40
3221	20.457722400	2404:6800:4009:82b::...	2405:201:5017:7886::...	UDP	88	443 → 65051 Len=26
3223	20.588198500	2405:201:5017:7886::...	2404:6800:4013:807::...	QUIC	486	Protected Payload (KP0), DCID=fac641bab8991618
3228	20.630965700	2404:6800:4013:807::...	2405:201:5017:7886::...	QUIC	94	Protected Payload (KP0)
3229	20.666102800	2405:201:5017:7886::...	2404:6800:4013:807::...	QUIC	94	Protected Payload (KP0), DCID=fac641bab8991618
3230	20.673250300	2404:6800:4013:807::...	2405:201:5017:7886::...	QUIC	156	Protected Payload (KP0)
3231	20.673250300	2404:6800:4013:807::...	2405:201:5017:7886::...	QUIC	87	Protected Payload (KP0)
3232	20.674094700	2405:201:5017:7886::...	2404:6800:4013:807::...	QUIC	97	Protected Payload (KP0), DCID=fac641bab8991618
3235	20.713440100	2404:6800:4013:807::...	2405:201:5017:7886::...	QUIC	90	Protected Payload (KP0)
3241	21.401575900	2404:6800:4002:808::...	2405:201:5017:7886::...	UDP	143	443 → 59828 Len=77
3242	21.434790400	2405:201:5017:7886::...	2404:6800:4002:808::...	UDP	94	59828 → 443 Len=32
3251	24.050528900	2405:201:5017:7886::...	2405:200:161c:1725::...	UDP	1288	58166 → 443 Len=1226
3252	24.050609600	2405:201:5017:7886::...	2405:200:161c:1725::...	UDP	1292	58166 → 443 Len=1230
3253	24.050653200	2405:201:5017:7886::...	2405:200:161c:1725::...	UDP	835	58166 → 443 Len=773
3254	24.062964000	2405:200:161c:1725::...	2405:201:5017:7886::...	UDP	186	443 → 58166 Len=124
3255	24.063386600	2405:200:161c:1725::...	2405:201:5017:7886::...	UDP	1288	443 → 58166 Len=1226
3256	24.063628800	2405:200:161c:1725::...	2405:201:5017:7886::...	UDP	1292	443 → 58166 Len=1230
3257	24.064000000	2405:200:161c:1725::...	2405:201:5017:7886::...	UDP	1292	443 → 58166 Len=1230
3258	24.064000000	2405:200:161c:1725::...	2405:201:5017:7886::...	UDP	1292	443 → 58166 Len=1230

No.	Time	Source	Destination	Protocol	Length	Info
78	1.337759600	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	167	Standard query response 0xc0b2 A api.x.com CNAME dynamic.x.com.cdn.cloudflare.net A 172.66.0.227 A 162.159.140.229
95	1.345796600	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	190	Standard query response 0xa7bc HTTPS api.x.com CNAME dynamic.x.com.cdn.cloudflare.net SOA ns1.cloudflare.net
3131	10.189165900	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	114	Standard query 0x3a32 A p2p-maa2.discovery.steamserver.net
3132	10.206874300	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	146	Standard query response 0x3a32 A p2p-maa2.discovery.steamserver.net A 155.133.225.21 A 155.133.225.20
3202	20.301843800	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	96	Standard query 0x805e HTTPS gator.volces.com
3203	20.302033900	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	96	Standard query 0x640c AAAA gator.volces.com
3204	20.302179900	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	96	Standard query 0xc096 A gator.volces.com
3213	20.327514800	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	321	Standard query response 0xc096 A gator.volces.com CNAME gator.volces.com.bytedns1.com CNAME gator.volces.com.230b2a2545cfa773...
3214	20.328467200	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	251	Standard query response 0x805e HTTPS gator.volces.com CNAME gator.volces.com.bytedns1.com CNAME gator.volces.com.230b2a2545cfa7...
3215	20.328467200	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	251	Standard query response 0x640c AAAA gator.volces.com CNAME gator.volces.com.bytedns1.com CNAME gator.volces.com.230b2a2545cfa7...
7459	26.259638400	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	96	Standard query 0xc0b8 HTTPS web.whatsapp.com
7460	26.259802400	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	96	Standard query 0x0754 AAAA web.whatsapp.com
7461	26.259931700	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	96	Standard query 0x8a54 A web.whatsapp.com
7462	26.273864500	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	178	Standard query response 0xc0b8 HTTPS web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net SOA a.ns.whatsapp.net
7463	26.276854900	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	161	Standard query response 0x0754 AAAA web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net AAAA 2a03:2880:f349:120:face:b00c:0:167
7464	26.278525000	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	149	Standard query response 0x8a54 A web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net A 57.144.49.32
7502	27.246672600	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	95	Standard query 0x4f1c HTTPS ssl.gstatic.com
7503	27.246876800	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	95	Standard query 0x88d1 AAAA ssl.gstatic.com
7504	27.247014000	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	95	Standard query 0xc092 A ssl.gstatic.com
7505	27.255103000	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	123	Standard query response 0x88d1 AAAA ssl.gstatic.com AAAA 2404:6800:4002:805::2003

Analyze a Packet

Wireshark - Packet 4 - Wi-Fi	
▶	Frame 4: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on inter
▶	Ethernet II, Src: Intel_85:54:30 (d0:ab:d5:85:54:30), Dst: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4)
▼	Internet Protocol Version 6, Src: 2405:201:5017:7886:c8bc:c42e:6da3:bb13, Dst: 2405:200:161c:1725::443
0110 = Version: 6	
.... 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: N	
.... 0000 00 = Differentiated Services Codepoint: 0	
.... 00 = Explicit Congestion Notification: N	
.... 0010 0001 1010 0010 = Flow Label: 0x221a2	
Payload Length: 20	
Next Header: TCP (6)	
Hop Limit: 64	
▶ Source Address: 2405:201:5017:7886:c8bc:c42e:6da3:bb13	
▶ Destination Address: 2405:200:161c:b9e::10	
[Stream index: 1]	
▶ Transmission Control Protocol, Src Port: 60600, Dst Port: 443, Seq: 1, Ack: 2, Len: 0	
0000 8c 83 94 c8 51 f4 d0 ab d5 85 54 30 86 dd 60 02	...Q...T0...
0010 21 a2 00 14 06 40 24 05 02 01 50 17 78 86 c8 bc	!...@\$.P.x...
0020 c4 2e 6d a3 bb 13 24 05 02 00 16 30 0b 9e 00 00	..m...\$...0...
0030 00 00 00 00 00 10 ec b8 01 bb 25 c8 c5 94 57 92	%...W...
0040 c6 40 50 10 00 ff ec 43 00 00	@P...C...

- Observe:
 - Ethernet header-

```

Ethernet II, Src: Intel_85:54:30 (d0:ab:d5:85:54:30), Dst: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4)
▶ Destination: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4)
▶ Source: Intel_85:54:30 (d0:ab:d5:85:54:30)
Type: IPv6 (0x86dd)
[Stream index: 0]

```

○ IP header-

```
Internet Protocol Version 6, Src: 2405:201:5017:7886:c8bc:c42e:6da3:bb13, Dst: 2405:200:1630:b9e::10
0110 .... = Version: 6
▼ .... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
    .... 0000 00.. .... = Differentiated Services Codepoint: Default (0)
    .... ..00 .... = Explicit Congestion Notification: Not ECN-Capable Transp
.... 0010 0010 0001 1010 0010 = Flow Label: 0x221a2
Payload Length: 20
Next Header: TCP (6)
Hop Limit: 64
```

○ TCP/UDP header-

```
▼ Transmission Control Protocol, Src Port: 60600, Dst Port: 443, Seq: 1, Ack: 2, Len: 0
Source Port: 60600
Destination Port: 443
[Stream index: 0]
[Stream Packet Number: 2]
▼ [Conversation completeness: Incomplete (60)]
    ..1. .... = RST: Present
    ...1 .... = FIN: Present
    .... 1... = Data: Present
    .... .1.. = ACK: Present
    .... ..0. = SYN-ACK: Absent
    .... ...0 = SYN: Absent
    [Completeness Flags: RFDA..]
    [TCP Segment Len: 0]
    Sequence Number: 1 (relative sequence number)
    Sequence Number (raw): 633914772
    [Next Sequence Number: 1 (relative sequence number)]
    Acknowledgment Number: 2 (relative ack number)
    Acknowledgment number (raw): 1469236800
    0101 .... = Header Length: 20 bytes (5)
▶ Flags: 0x010 (ACK)
Window: 255
[Calculated window size: 255]
[Window size scaling factor: -1 (unknown)]
Checksum: 0xec43 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
▶ [Timestamps]
▶ [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
```

○ Payload data-

0000	8c 83 94 c8 51 f4 d0 ab d5 85 54 30 86 dd 60 02	...Q...T0..`.
0010	21 a2 00 14 06 40 24 05 02 01 50 17 78 86 c8 bc	!...@\$..P.x...
0020	c4 2e 6d a3 bb 13 24 05 02 00 16 30 0b 9e 00 00	..m...\$. ...0....
0030	00 00 00 00 00 10 ec b8 01 bb 25 c8 c5 94 57 92	%...W.
0040	c6 40 50 10 00 ff ec 43 00 00	..@P...C ..

Observation

- Interface used: - Wireshark
- Protocols observed: TCP
- Source and destination IP addresses:

Source: 2405:201:5017:7886:c8bc:c42e:6da3:bb13

Destination: 2405:200:1630:b9e::10

- Packet length: 74 bytes

Result

Live network packets were successfully captured using Wireshark, and basic network protocols were identified and analyzed.

ASSIGNMENT 2

1. To capture and analyze ICMP packets (Ping request and reply)

6399	54.976377500	192.168.29.11	8.8.8.8	ICMP	74 Echo (ping) request	id=0x0001, seq=1/256, ttl=128 (reply in 6400)
6400	55.205798200	8.8.8.8	192.168.29.11	ICMP	78 Echo (ping) reply	id=0x0001, seq=1/256, ttl=112 (request in 6399)
6401	55.410151500	18.66.78.127	192.168.29.11	TLSv1.2	176 Application Data	
6402	55.453219700	192.168.29.11	18.66.78.127	TCP	54 64249 → 443 [ACK] Seq=29 Ack=758 Win=1020 Len=0	
6403	55.997775100	192.168.29.11	8.8.8.8	ICMP	74 Echo (ping) request	id=0x0001, seq=2/512, ttl=128 (reply in 6404)
6404	56.016254100	8.8.8.8	192.168.29.11	ICMP	78 Echo (ping) reply	id=0x0001, seq=2/512, ttl=112 (request in 6403)
6405	57.020175900	192.168.29.11	8.8.8.8	ICMP	74 Echo (ping) request	id=0x0001, seq=3/768, ttl=128 (reply in 6406)
6406	57.036305400	8.8.8.8	192.168.29.11	ICMP	78 Echo (ping) reply	id=0x0001, seq=3/768, ttl=112 (request in 6405)
6407	58.041787100	192.168.29.11	8.8.8.8	ICMP	74 Echo (ping) request	id=0x0001, seq=4/1024, ttl=128 (reply in 6408)
6408	58.059955100	8.8.8.8	192.168.29.11	ICMP	78 Echo (ping) reply	id=0x0001, seq=4/1024, ttl=112 (request in 6407)
6409	58.572095500	2405:201:5017:7886::...	2a03:2880:f349:120::...	TLSv1.2	144 Application Data	
6410	58.583132200	192.168.29.1	224.0.0.1	IGMPv3	50 Membership Query, general	
6411	58.583132200	fe80::8e83:94ff:fec::...	ff02::1	ICMPv6	90 Multicast Listener Query	
6412	58.583132200	fe80::8e83:94ff:fec::...	ff02::1	ICMPv6	90 Multicast Listener Query	
6413	58.583387300	fe80::2641:a5cd:c69::...	ff02::16	ICMPv6	90 Multicast Listener Report Message v2	

Request ICMP Packet:

```
Frame 6403: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{86F0A4D7-7347-4067-B9BB-6C9A9DEEC237}, id 0
Ethernet II, Src: Intel_85:54:30 (d0:ab:d5:85:54:30), Dst: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4)
Internet Protocol Version 4, Src: 192.168.29.11, Dst: 8.8.8.8
Internet Control Message Protocol

0000  8c 83 94 c8 51 f4 d0 ab d5 85 54 30 08 00 45 00  ....Q...T..E.
0010  00 3c 9f 0c 00 00 80 01 00 00 c0 a8 1d 0b 08 08  -<.....p....
0020  08 08 08 00 4d 59 00 01 00 02 61 62 63 64 65 66  -MY....abcdef
0030  67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76  ghijklmn opqrstuv
```

Reply ICMP Packet:

```
Frame 6404: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{86F0A4D7-7347-4067-B9BB-6C9A9DEEC237}, id 0
Ethernet II, Src: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4), Dst: Intel_85:54:30 (d0:ab:d5:85:54:30)
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
Internet Protocol Version 4, Src: 8.8.8.8, Dst: 192.168.29.11
Internet Control Message Protocol

0000  d0 ab d5 85 54 30 8c 83 94 c8 51 f4 81 00 00 00  -...T...Q....
0010  08 00 45 80 00 3c 00 00 00 00 70 01 5c 7e 08 08  -E...<...p...
0020  08 08 0c a8 1d 0b 00 00 55 59 00 01 00 02 61 62  -UY....ab....
0030  63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72  cdefghij klmnopqr
0040  73 74 75 76 77 61 62 63 64 65 66 67 68 69      stuvwabc defghi
```

2. To analyze TCP packets and observe the TCP three-way handshake

5643	250.378756500	192.168.29.11	18.236.36.28	TCP	66 49170 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
5644	250.420362800	34.218.221.118	192.168.29.11	TCP	54 80 → 50464 [ACK] Seq=9909 Ack=546 Win=28160 Len=0	
5645	250.430421200	34.218.221.118	192.168.29.11	TCP	54 80 → 62410 [ACK] Seq=2 Ack=2 Win=27136 Len=0	
5646	250.431861900	34.218.221.118	192.168.29.11	TCP	54 80 → 60887 [ACK] Seq=2 Ack=2 Win=27136 Len=0	
5647	250.443406400	18.236.36.28	192.168.29.11	TCP	66 80 → 52358 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1460 SACK_PERM WS=256	
5648	250.443474000	192.168.29.11	18.236.36.28	TCP	54 52358 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0	
5649	250.443794100	192.168.29.11	18.236.36.28	HTTP	624 GET /testcat_ac.html HTTP/1.1	
5650	250.445283500	18.236.36.28	192.168.29.11	TCP	66 80 → 54271 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1460 SACK_PERM WS=256	
5651	250.445318800	192.168.29.11	18.236.36.28	TCP	54 54271 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0	
5652	250.620453000	18.66.78.127	192.168.29.11	TLSv1.2	177 Application Data	
5653	250.662561000	192.168.29.11	18.66.78.127	TCP	54 64249 → 443 [ACK] Seq=113 Ack=3269 Win=1023 Len=0	
5654	250.666538600	18.236.36.28	192.168.29.11	TCP	66 80 → 49170 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1460 SACK_PERM WS=256	
5655	250.666613900	192.168.29.11	18.236.36.28	TCP	54 49170 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0	
5656	250.735952000	2404:6800:4002:810::...	2405:201:5017:7886::...	UDP	261 443 → 62996 Len=195	
5657	250.735952000	18.236.36.28	192.168.29.11	TCP	54 80 → 52358 [ACK] Seq=1 Ack=571 Win=28160 Len=0	
5658	250.739079300	18.236.36.28	192.168.29.11	TCP	1514 80 → 52358 [ACK] Seq=1 Ack=571 Win=28160 Len=1460 [TCP PDU reassembled in 5668]	
5659	250.739217700	2405:201:5017:7886::...	2404:6800:4002:810::...	UDP	95 62996 → 443 Len=33	
5660	250.740222200	18.236.36.28	192.168.29.11	TCP	1514 80 → 52358 [ACK] Seq=1461 Ack=571 Win=28160 Len=1460 [TCP PDU reassembled in 5668]	
5661	250.740256300	192.168.29.11	18.236.36.28	TCP	54 52358 → 80 [ACK] Seq=571 Ack=2921 Win=65280 Len=0	
5662	250.741352700	18.236.36.28	192.168.29.11	TCP	1514 80 → 52358 [ACK] Seq=2921 Ack=571 Win=28160 Len=1460 [TCP PDU reassembled in 5668]	
5663	250.744703100	18.236.36.28	192.168.29.11	TCP	1514 80 → 52358 [ACK] Seq=4381 Ack=571 Win=28160 Len=1460 [TCP PDU reassembled in 5668]	
5664	250.744797200	192.168.29.11	18.236.36.28	TCP	54 52358 → 80 [ACK] Seq=571 Ack=5841 Win=65280 Len=0	
5665	250.745517900	18.236.36.28	192.168.29.11	TCP	1514 80 → 52358 [ACK] Seq=5841 Ack=571 Win=28160 Len=1460 [TCP PDU reassembled in 5668]	
5666	250.747292100	18.236.36.28	192.168.29.11	TCP	1514 80 → 52358 [ACK] Seq=7301 Ack=571 Win=28160 Len=1460 [TCP PDU reassembled in 5668]	
5667	250.747370400	192.168.29.11	18.236.36.28	TCP	54 52358 → 80 [ACK] Seq=571 Ack=8761 Win=65280 Len=0	
5668	250.747612600	192.168.29.11	18.236.36.28	HTTP	1201 HTTP/1.1 200 OK (text/html)	

SYN Packet:

```
Frame 5643: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{86F0A4D7-7347-4067-B9BB-6C9A9DEEC237}, id 0
Ethernet II, Src: Intel_85:54:30 (d0:ab:d5:85:54:30), Dst: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4)
Internet Protocol Version 4, Src: 192.168.29.11, Dst: 18.236.36.28
Transmission Control Protocol, Src Port: 49170, Dst Port: 80, Seq: 0, Len: 0

0000  8c 83 94 c8 51 f4 d0 ab d5 85 54 30 08 00 45 00  ....Q...T0.E
0010  00 34 30 11 40 00 80 06 00 00 c0 a8 1d 0b 12 ec  40 @ .....
0020  24 1c c0 12 00 50 d0 19 9a ba 00 00 00 80 02    $.-P.....
0030  ff ff 14 e2 00 00 02 04 05 b4 01 03 03 08 01 01  .....
0040  04 02  ..
```

SYN-ACK Packet

```
Frame 5647: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{86F0A4D7-7347-4067-B9BB-6C9A9DEEC237}, id 0
Ethernet II, Src: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4), Dst: Intel_85:54:30 (d0:ab:d5:85:54:30)
Internet Protocol Version 4, Src: 18.236.36.28, Dst: 192.168.29.11
Transmission Control Protocol, Src Port: 80, Dst Port: 52358, Seq: 0, Ack: 1, Len: 0

0000  d0 ab d5 85 54 30 8c 83 94 c8 51 f4 08 00 45 00  ....T0...Q...E
0010  00 34 00 00 40 00 ef 06 77 08 12 ec 24 1c c0 a8  4 @ ...w...$...
0020  1d 0b 00 50 cc 86 89 82 15 fd d3 9c 29 5e 80 12  ..P.....)^...
0030  69 03 87 f0 00 00 02 04 05 b4 01 01 04 02 01 03  i.....
0040  03 08  ..
```

ACK Packet

```
Frame 5648: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{86F0A4D7-7347-4067-B9BB-6C9A9DEEC237}, id 0
Ethernet II, Src: Intel_85:54:30 (d0:ab:d5:85:54:30), Dst: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4)
Internet Protocol Version 4, Src: 192.168.29.11, Dst: 18.236.36.28
Transmission Control Protocol, Src Port: 52358, Dst Port: 80, Seq: 1, Ack: 1, Len: 0

0000  8c 83 94 c8 51 f4 d0 ab d5 85 54 30 08 00 45 00  ....Q...T0.E
0010  00 28 30 12 40 00 80 06 00 00 c0 a8 1d 0b 12 ec  0 @ .....
0020  24 1c cc 86 00 50 d3 9c 29 5e 89 82 15 fe 50 10  $.-P...)^...P
0030  00 ff 14 d6 00 00  .....

```

3. To capture and analyze UDP packets

5656	250.735852000	2404:6800:4002:810::	2405:201:5017:7886::	UDP	261 443 → 62996 Len=199
5659	250.739217700	2405:201:5017:7886::	2404:6800:4002:810::	UDP	95 62996 → 443 Len=33
5669	250.773408400	2405:201:5017:7886::	2405:201:5017:7886::	DNS	94 Standard query 0x1602 HTTPS www.google.com
5670	250.773742300	2405:201:5017:7886::	2405:201:5017:7886::	DNS	94 Standard query 0x3c83 AAAA www.google.com
5671	250.773995700	2405:201:5017:7886::	2405:201:5017:7886::	DNS	94 Standard query 0x922d A www.google.com
5672	250.782300700	2405:201:5017:7886::	2405:201:5017:7886::	DNS	318 Standard query response 0x3c83 AAAA www.google.com AAAA 2001:4860:482d:7700::
5673	250.785483900	2405:201:5017:7886::	2405:201:5017:7886::	DNS	222 Standard query response 0x922d A www.google.com A 142.251.151.119 A 142.251.151.119
5674	250.787495200	2405:201:5017:7886::	2405:201:5017:7886::	DNS	119 Standard query response 0x1602 HTTPS www.google.com HTTPS
5676	250.808817400	2405:201:5017:7886::	2001:4860:482a:7700::	UDP	91 65435 → 443 Len=29
5677	250.827516100	2001:4860:482a:7700::	2405:201:5017:7886::	UDP	88 443 → 65435 Len=26
5678	250.891131300	2405:201:5017:7886::	2404:6800:4002:810::	UDP	499 62996 → 443 Len=437
5679	250.907334300	2404:6800:4002:810::	2405:201:5017:7886::	UDP	92 443 → 62996 Len=30
5680	250.913827800	2405:201:5017:7886::	2404:6800:4002:810::	UDP	95 62996 → 443 Len=33
5681	251.012932000	2405:201:5017:7886::	2606:4700:3033::ac4::	QUIC	1292 Initial, DCID=bccbebf51d4b35c, PKN: 1, PADDING, CRYPTO, CRYPTO, CRYPTO, CRYPTO
5682	251.013010900	2405:201:5017:7886::	2606:4700:3033::ac4::	QUIC	1292 Initial, DCID=bccbebf51d4b35c, PKN: 2, CRYPTO, PADDING, CRYPTO, PING, PADDING
5683	251.235662800	2404:6800:4002:810::	2405:201:5017:7886::	UDP	1072 443 → 62996 Len=1010
5684	251.235662800	2404:6800:4002:810::	2405:201:5017:7886::	UDP	85 443 → 62996 Len=23
5685	251.235662800	2606:4700:3033::ac4::	2405:201:5017:7886::	QUIC	1262 Initial, SCID=01c708ed50bb5d7065c524ed6db97a7065c4268c, PKN: 0, ACK, CRYPTO
5686	251.235662800	2606:4700:3033::ac4::	2405:201:5017:7886::	QUIC	1262 Handshake, SCID=01c708ed50bb5d7065c524ed6db97a7065c4268c
5687	251.236344200	2405:201:5017:7886::	2606:4700:3033::ac4::	QUIC	1292 Protected Payload (KP0), DCID=01c708ed50bb5d7065c524ed6db97a7065c4268c
5688	251.236717700	2405:201:5017:7886::	2606:4700:3033::ac4::	QUIC	1286 Protected Payload (KP0), DCID=01c708ed50bb5d7065c524ed6db97a7065c4268c
5689	251.236755900	2405:201:5017:7886::	2606:4700:3033::ac4::	QUIC	442 Protected Payload (KP0), DCID=01c708ed50bb5d7065c524ed6db97a7065c4268c
5690	251.239539100	2405:201:5017:7886::	2404:6800:4002:810::	UDP	99 62996 → 443 Len=37
5691	251.255435000	2404:6800:4002:810::	2405:201:5017:7886::	UDP	88 443 → 62996 Len=26

UDP Packet:

```
▶ Frame 1822: Packet, 176 bytes on wire (1408 bits), 176 bytes captured (1408 bits) on interface \Device\NPF_{86...}
▶ Ethernet II, Src: 8a:12:ad:70:e0:69 (8a:12:ad:70:e0:69), Dst: Intel_85:54:30 (d0:ab:d5:85:54:30)
▶ Internet Protocol Version 6, Src: 2405:201:5017:7886::c0a8:1d01, Dst: 2405:201:5017:7886:c8bc:c42e:6da3:bb13
▶ User Datagram Protocol, Src Port: 53, Dst Port: 57389
▶ Domain Name System (response)

0000  d0 ab d5 85 54 30 8a 12  ad 70 e0 69 86 dd 60 00  ...T0... .p.i...
0010  00 00 00 7a 11 40 24 05  02 01 50 17 78 86 00 00  ...z @$... .P.x...
0020  00 00 c0 a8 1d 01 24 05  02 01 50 17 78 86 c8 bc  ....@$... .P.x...
0030  c4 2e 6d a3 bb 13 00 35  e0 2d 00 7a 53 fb cd 7c  .m....5... .zS...|
0040  81 80 00 01 00 00 00 01  00 00 08 6e 65 76 65 72  ....never
0050  73 73 6c 03 63 6f 6d 00  00 41 00 01 c0 0c 00 06  ssl.com... .A.....
0060  00 01 00 00 03 84 00 48  07 6e 73 2d 31 37 31 36  ....H ns-1716
0070  09 61 77 73 64 6e 73 2d  32 32 02 63 6f 02 75 6b  .awsdns- 22 co uk
0080  00 11 61 77 73 64 6e 73  2d 68 6f 73 74 6d 61 73  .awsdns -hostmas
0090  74 65 72 06 61 6d 61 7a  6f 6e c0 15 00 00 00 01  ter amaz on.....
00a0  00 00 1c 20 00 00 03 84  00 12 75 00 00 01 51 80  ... ..u...Q.
```

4. To capture and analyze DNS query and response packets

226	10.535547900	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	155	Standard query response 0x20f9 HTTPS e2c39.gcp.gvt2.com SOA ns1.google.com
227	10.541800500	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	114	Standard query response 0xec1a A e2c39.gcp.gvt2.com A 35.217.17.196
228	10.543376200	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	155	Standard query response 0xdb58 AAAA e2c39.gcp.gvt2.com SOA ns1.google.com
2003	15.905653300	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	136	Standard query 0x9206 HTTPS prod-dynamite-prod-00-us-signaler-pa.clients6.google.com
2004	15.905849000	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	136	Standard query 0x27fc AAAA prod-dynamite-prod-00-us-signaler-pa.clients6.google.com
2005	15.906006400	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	136	Standard query 0x4dc2 A prod-dynamite-prod-00-us-signaler-pa.clients6.google.com
2007	15.925479500	2405:201:5017:7886::...	2405:201:5017:7886::...	DNS	186	Standard query response 0x9206 HTTPS prod-dynamite-prod-00-us-signaler-pa.clients6.google.com SOA ns1.google.com

DNS Packet:

```
▶ Frame 1820: Packet, 108 bytes on wire (864 bits), 108 bytes captured (864 bits) on interface \Device\NPF_{86F0A4D7-7...}
▶ Ethernet II, Src: 8a:12:ad:70:e0:69 (8a:12:ad:70:e0:69), Dst: Intel_85:54:30 (d0:ab:d5:85:54:30)
▶ Internet Protocol Version 6, Src: 2405:201:5017:7886::c0a8:1d01, Dst: 2405:201:5017:7886:c8bc:c42e:6da3:bb13
▶ User Datagram Protocol, Src Port: 53, Dst Port: 54965
▶ Domain Name System (response)

0000  d0 ab d5 85 54 30 8a 12  ad 70 e0 69 86 dd 60 00  ...T0... .p.i...
0010  00 00 00 36 11 40 24 05  02 01 50 17 78 86 00 00  ...6 @$... .P.x...
0020  00 00 c0 a8 1d 01 24 05  02 01 50 17 78 86 c8 bc  ....@$... .P.x...
0030  c4 2e 6d a3 bb 13 00 35  d6 b5 00 36 4c 2a 07 f7  .m....5... .6L*...
0040  81 80 00 01 00 00 00 00  00 00 08 6e 65 76 65 72  ....never
0050  73 73 6c 03 63 6f 6d 00  00 01 00 01 c0 0c 00 01  ssl.com... ..
0060  00 01 00 00 00 01 00 04  22 df 7c 2d              .... " |-
```

5. To capture and analyze HTTP packets

No.	Time	Source	Destination	Protocol	Length Info
1874	4.123211800	192.168.29.11	34.223.124.45	HTTP	514 GET / HTTP/1.1
1890	4.711416300	34.223.124.45	192.168.29.11	HTTP	867 HTTP/1.1 200 OK (text/html)

HTTP Packet:

```
▶ Frame 1874: Packet, 514 bytes on wire (4112 bits), 514 bytes captured (4112 bits) on interface \Device\NPF_{86F0A4D7-7347-4067-B9BB-6C9A9DEEC237}, id 0
▶ Ethernet II, Src: Intel_85:54:30 (d0:ab:d5:85:54:30), Dst: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4)
▶ Internet Protocol Version 4, Src: 192.168.29.11, Dst: 34.223.124.45
▶ Transmission Control Protocol, Src Port: 61961, Dst Port: 80, Seq: 1, Ack: 1, Len: 460
▶ Hypertext Transfer Protocol

0000  8c 83 94 c8 51 f4 d0 ab d5 85 54 30 08 00 45 00  ....Q...-T0..E.
0010  01 f4 90 09 40 00 80 06 00 00 c0 a8 1d 0b 22 df  ....@... ..".
0020  7c 2d f2 09 00 50 d7 f7 5a 16 e7 4c fc 36 50 18  |---P..Z..L6P.
0030  ff ff 7e a6 00 00 47 45 54 20 2f 20 48 54 54 50  ...~...GE T / HTTP
0040  2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 6e 65 76 65  /1.1..Host: neve
0050  72 73 73 6c 2e 63 6f 6d 0d 0a 43 6f 6e 6e 65 63  rssl.com ..Connec
0060  74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65  tion: ke ep-alive
0070  0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63 75  ..Upgrad e-Insecu
0080  72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 0d 0a  re-Reque sts: 1..
0090  55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69  User-Age nt: Mozi
00a0  6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73  lla/5.0 (Windows
00b0  20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36 34 3b  NT 10.0 ; Win64;
00c0  20 78 36 34 29 20 41 70 70 6c 65 57 65 62 4b 69  x64) Ap pleWebKi
00d0  74 2f 35 33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c  t/537.36 (KHTML,
00e0  20 6c 69 6b 65 20 47 65 63 6b 6f 29 20 43 68 72  like Ge cko) Chr
00f0  6f 6d 65 2f 31 34 37 2e 30 2e 30 2e 30 20 53 61  ome/147. 0.0.0 Sa
0100  66 61 72 69 2f 35 33 37 2e 33 36 0d 0a 41 63 63  fari/537 .36..Acc
0110  65 70 74 3a 20 74 65 78 74 2f 68 74 6d 6c 2c 61  ept: tex t/html,a
0120  70 70 6c 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c  pplicati on/xhtml
0130  2b 78 6d 6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e  +xml,app lication
0140  2f 78 6d 6c 3b 71 3d 30 2e 39 2c 69 6d 61 67 65  /xml;q=0 .9,image
0150  2f 61 76 69 66 2c 69 6d 61 67 65 2f 77 65 62 70  /avif,im age/webp
0160  2c 69 6d 61 67 65 2f 61 70 6e 67 2c 2a 2f 2a 3b  ,image/a png,*/*;
0170  71 3d 30 2e 38 2c 61 70 70 6c 69 63 61 74 69 6f  q=0.8,ap plicatio
0180  6e 2f 73 69 67 6e 65 64 2d 65 78 63 68 61 6e 67  n/signed -exchang
0190  65 3b 76 3d 62 33 3b 71 3d 30 2e 37 0d 0a 41 63  e;v=b3;q =0.7..Ac
01a0  63 65 70 74 2d 45 6e 63 6f 64 69 6e 67 3a 20 67  cept-Enc oding: g
01b0  7a 69 70 2c 20 64 65 66 6c 61 74 65 0d 0a 41 63  zip, def late ..Ac
01c0  63 65 70 74 2d 4c 61 6e 67 75 61 67 65 3a 20 65  cept-Lan guage: e
01d0  6e 2d 49 4e 2c 65 6e 2d 47 42 3b 71 3d 30 2e 39  n-IN,en- GB;q=0.9
01e0  2c 65 6e 2d 55 53 3b 71 3d 30 2e 38 2c 65 6e 3b  ,en-US;q =0.8,en;
01f0  71 3d 30 2e 37 2c 68 69 3b 71 3d 30 2e 36 0d 0a  q=0.7,hi ;q=0.6..
0200  0d 0a
```

6. To analyze ARP request and reply packets

No.	Time	Source	Destination	Protocol	Length	Info
1943	6.875228100	Arcadyan_c8:51:f4	Intel_85:54:30	ARP	42	Who has 192.168.29.11? Tell 192.168.29.11
1944	6.875242700	Intel_85:54:30	Arcadyan_c8:51:f4	ARP	42	192.168.29.11 is at d0:ab:d5:85:54:30

Request Packets:

```
▶ Frame 1943: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{86F0A4D7-7347-4067-B9BB-6C9A9DEEC237}, id 0
▶ Ethernet II, Src: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4), Dst: Intel_85:54:30 (d0:ab:d5:85:54:30)
▶ Address Resolution Protocol (request)

0000  d0 ab d5 85 54 30 8c 83 94 c8 51 f4 08 06 00 01  ....T0...-Q.....
0010  08 00 06 04 00 01 8c 83 94 c8 51 f4 c0 a8 1d 01  ....-Q.....
0020  00 00 00 00 00 00 c0 a8 1d 0b
```

Reply Packets:

```

> Frame 1944: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{86F0A4D7-7347-4067-B9BB-6C9A9DEEC237}, id 0
> Ethernet II, Src: Intel_85:54:30 (d0:ab:d5:85:54:30), Dst: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4)
> Address Resolution Protocol (reply)

0000  8c 83 94 c8 51 f4 d0 ab d5 85 54 30 08 06 00 01  ....Q...--T0...
0010  08 00 06 04 00 02 d0 ab d5 85 54 30 c0 a8 1d 0b  ....--T0...
0020  8c 83 94 c8 51 f4 c0 a8 1d 01  ....Q...--

```

7. To identify source and destination MAC and IP addresses in captured packets

```

> Frame 1990: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{86F0A4D7-7347-4067-B9BB-6C9A9DEEC237}, id 0
> Ethernet II, Src: Arcadyan_c8:51:f4 (8c:83:94:c8:51:f4), Dst: Intel_85:54:30 (d0:ab:d5:85:54:30)
> Internet Protocol Version 4, Src: 34.223.124.45, Dst: 192.168.29.11
> Transmission Control Protocol, Src Port: 80, Dst Port: 61961, Seq: 2274, Ack: 461, Len: 0

0000  d0 ab d5 85 54 30 8c 83 94 c8 51 f4 08 00 45 00  ....T0...Q...E.
0010  00 28 4f 9d 40 00 ef 06 bf 72 22 df 7c 2d c0 a8  (O@...r" |...
0020  1d 0b 00 50 f2 09 e7 4d 05 17 d7 f7 5b e2 50 11  ..P..M...[P.
0030  75 40 ab 3a 00 00  u@ :..

```

Source MAC address : 8c:83:94:c8:51:f4

Destination MAC address : d0:ab:d5:85:54:30

Source IP address : 34.223.124.45

Destination IP address 192.168.29.11

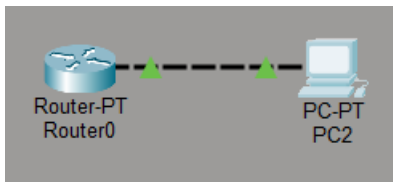
ASSIGNMENT 3

To create simple Network Topologies Using Cisco Packet Tracer.

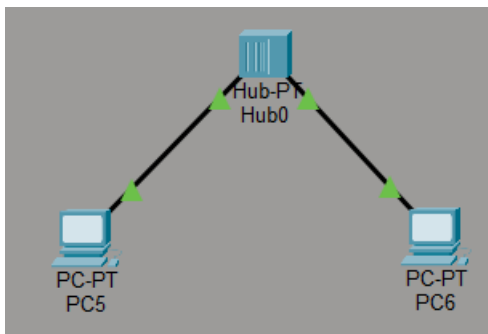
Q1.



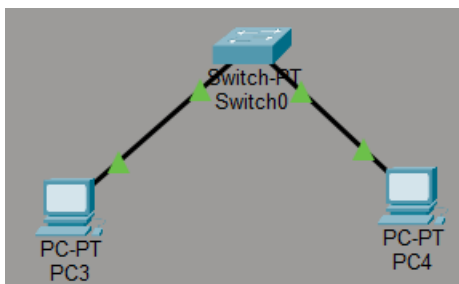
Q2.



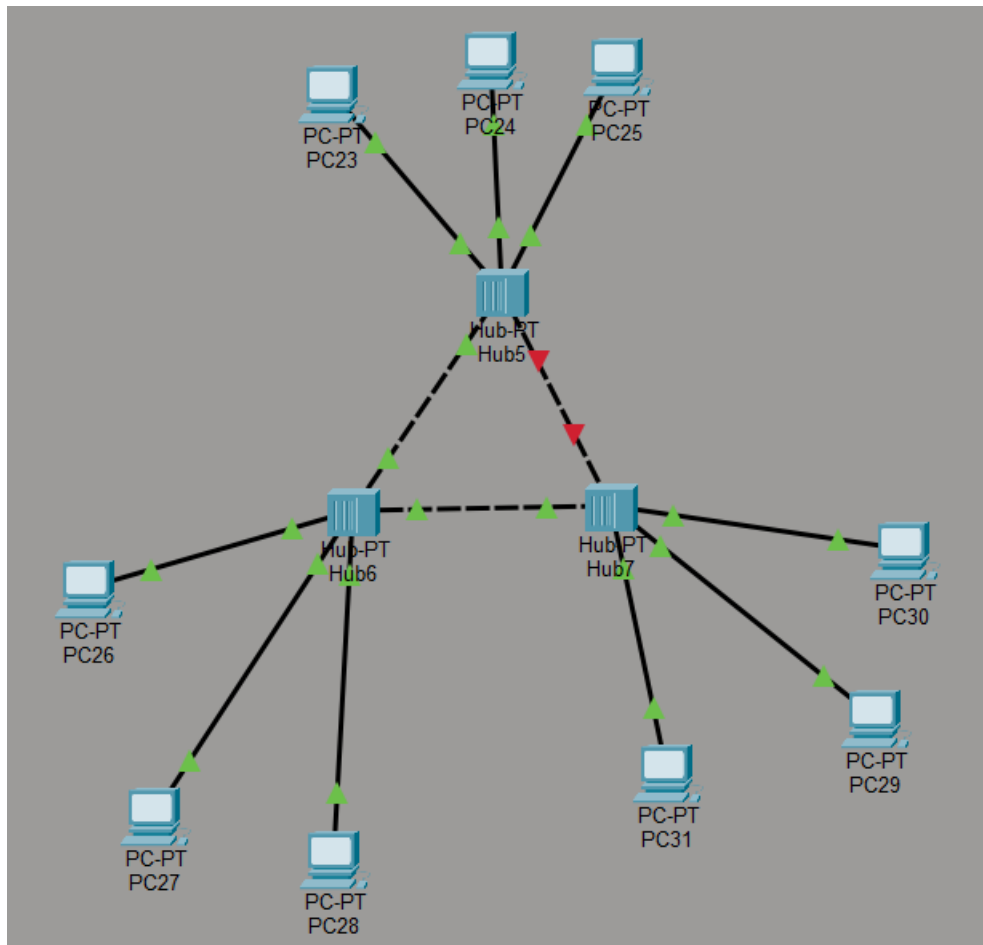
Q3.



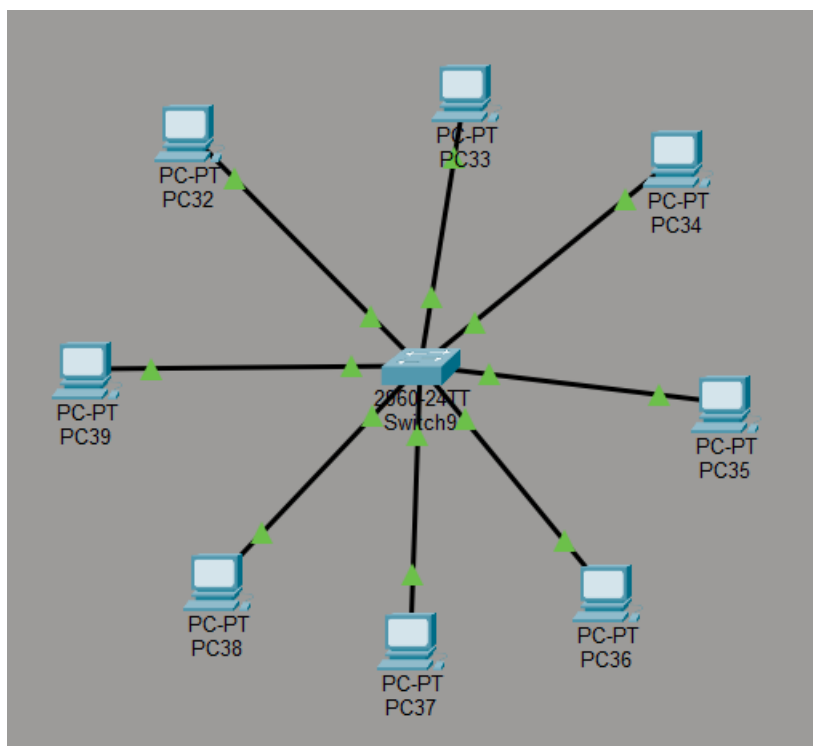
Q4.



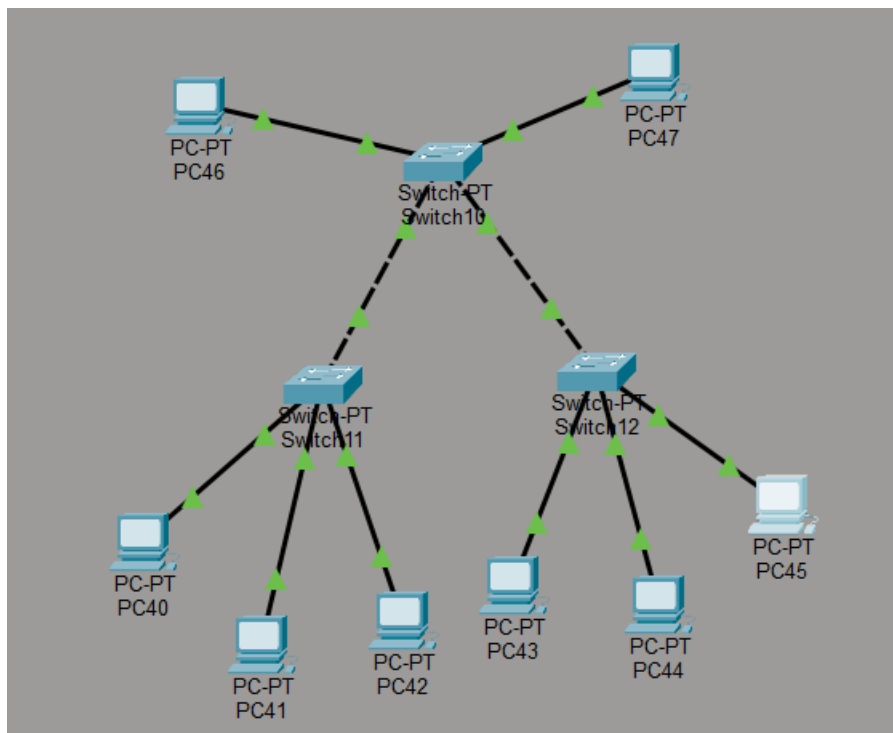
Q5.



Q7.



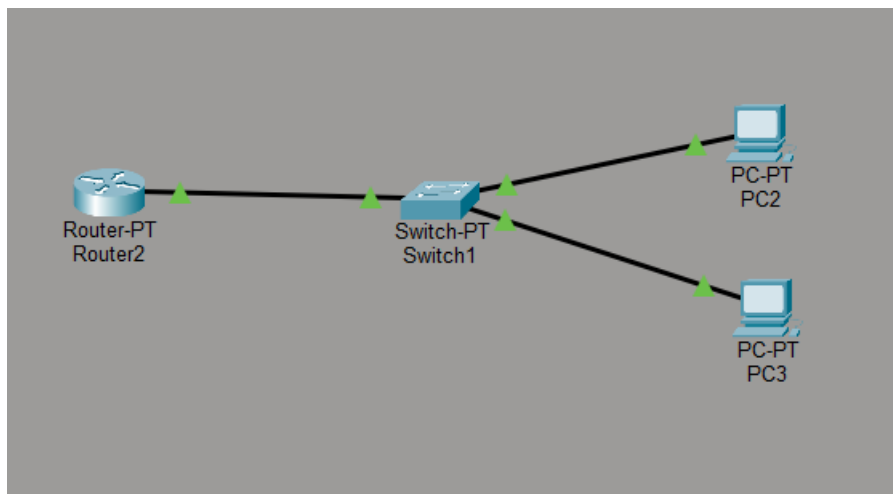
Q8.



ASSIGNMENT 4

Q.1 Create a pool on the DHCP Server router, named “xyz” and follow all the requirements.

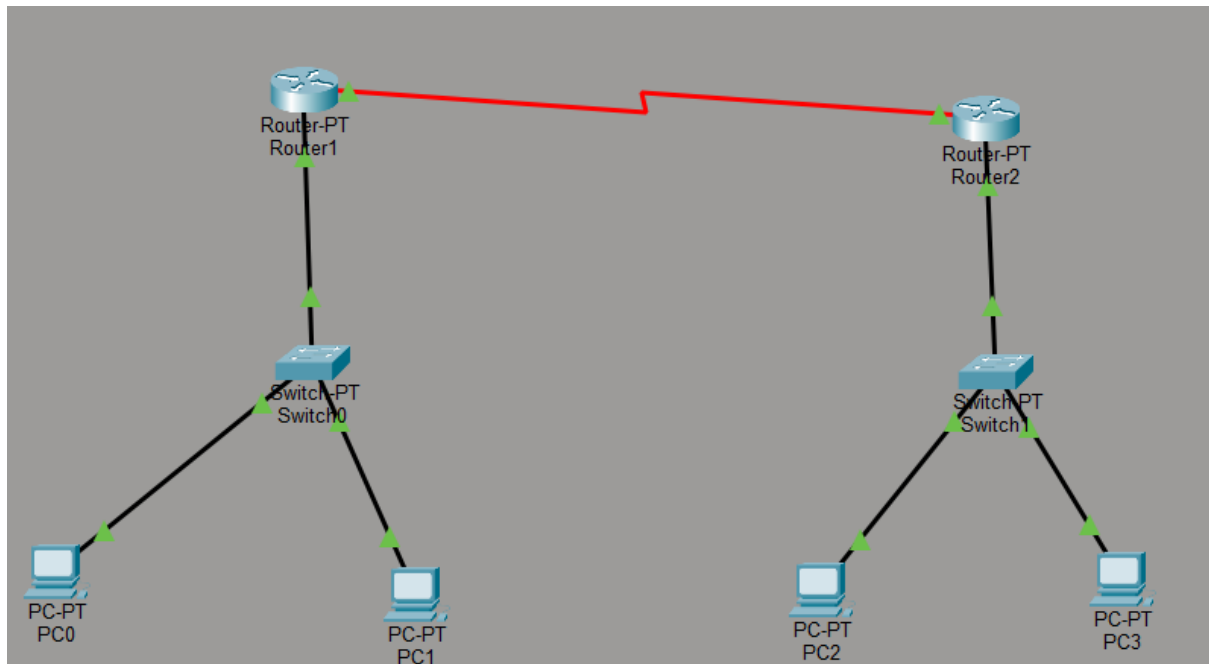
1. Client should use the DNS server with the IP address: 1.2.3.4
2. Client should use the network 192.168.10.0/24
3. Client should renew their IP address after every 3 hours
4. Make sure, Router does not respond to any BOOTP
5. Configure the Client so that it will obtain IP from DHCP server



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC3	PC2	ICMP		0.000	N	0	(edit)	

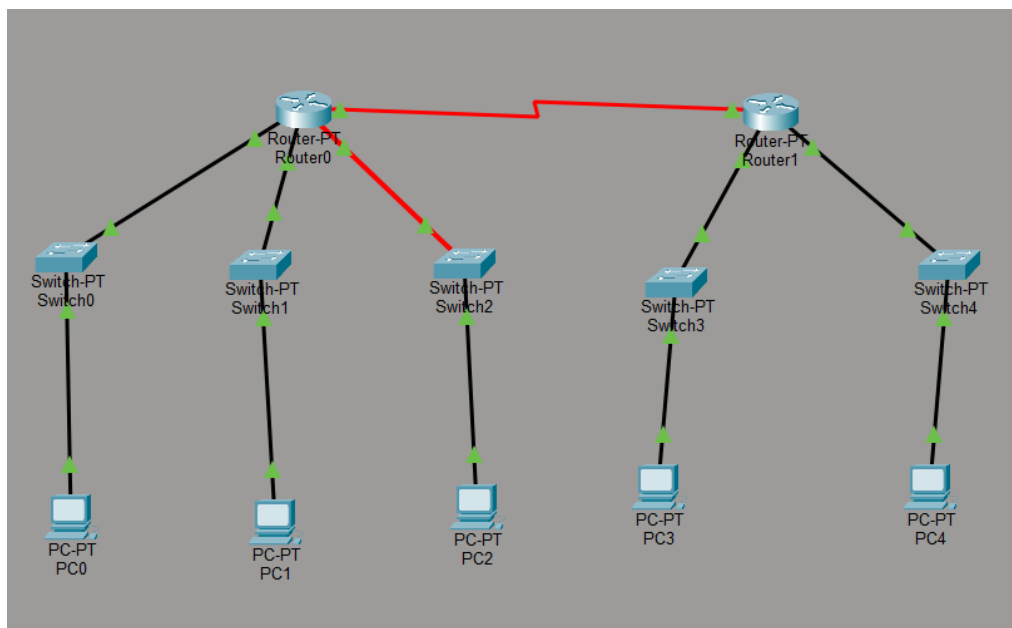
ASSIGNMENT 5

Q1.



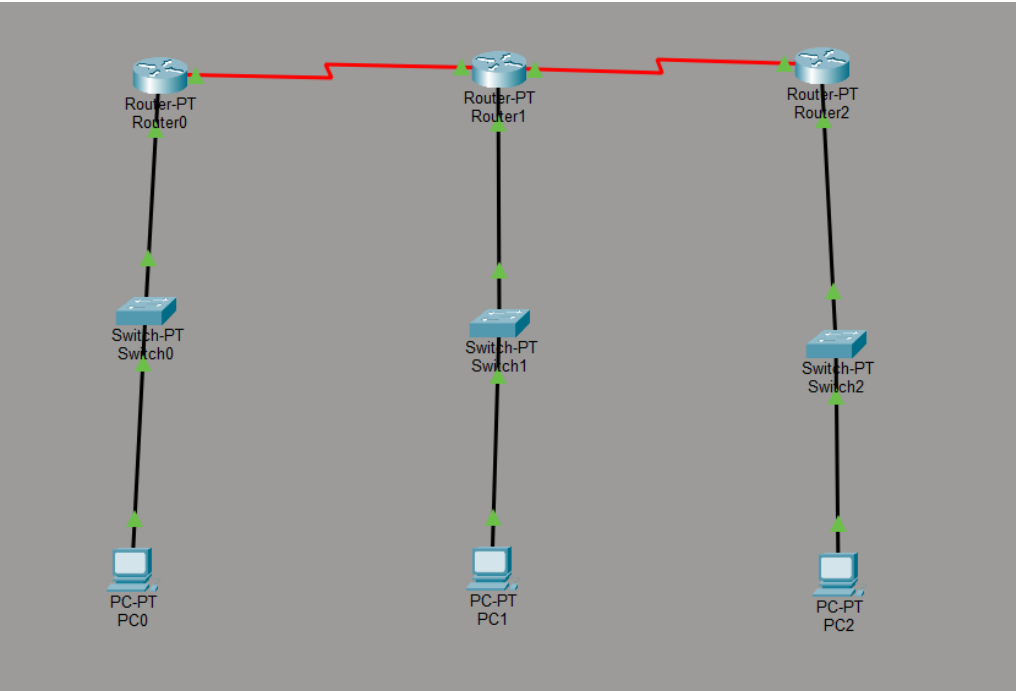
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP	Green	0.000	N	15	(edit)	
	Successful	PC0	PC2	ICMP	Green	0.000	N	16	(edit)	
	Failed	PC1	PC3	ICMP	Grey	0.000	N	17	(edit)	
	Successful	PC1	PC3	ICMP	Red	0.000	N	18	(edit)	

Q2.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	
	Successful	PC0	PC3	ICMP		0.000	N	1	(edit)	

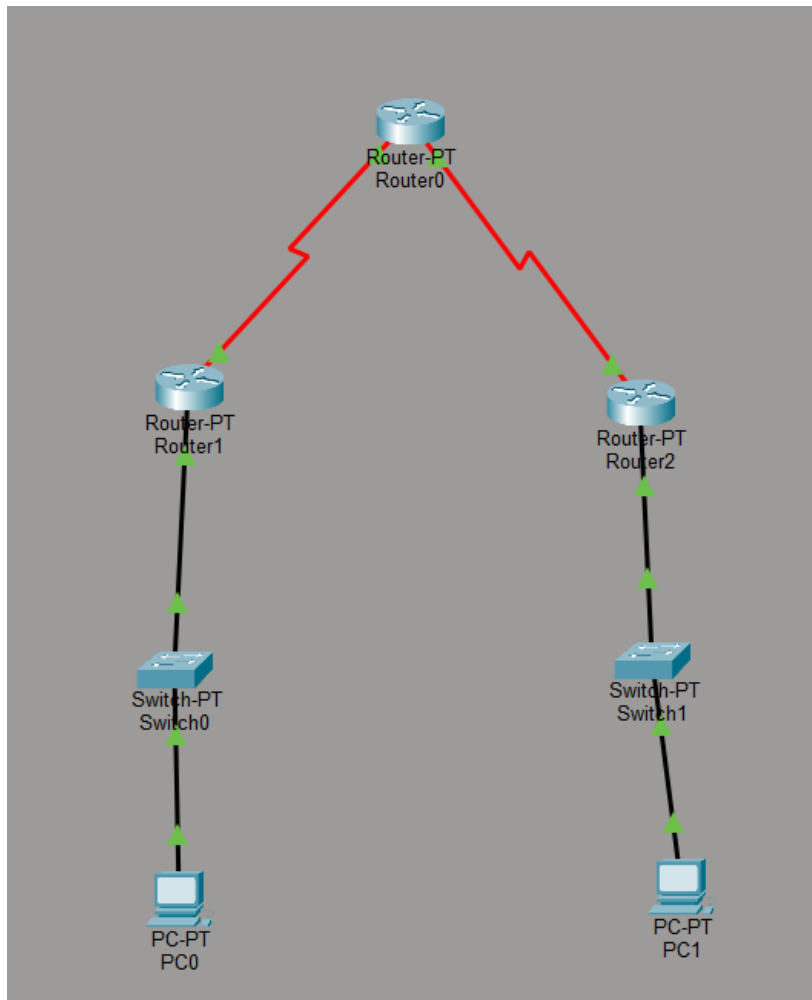
Q3.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC0	PC2	ICMP		0.000	N	2	(edit)	
	Successful	PC0	PC2	ICMP		0.000	N	3	(edit)	
	Failed	PC0	PC1	ICMP		0.000	N	4	(edit)	
	Successful	PC0	PC1	ICMP		0.000	N	5	(edit)	

ASSIGNMENT 6

To configure and verify routing using RIP Version 2 protocol in a network.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC0	PC1	ICMP		0.000	N	0	(edit)	
	Successful	PC0	PC1	ICMP		0.000	N	1	(edit)	

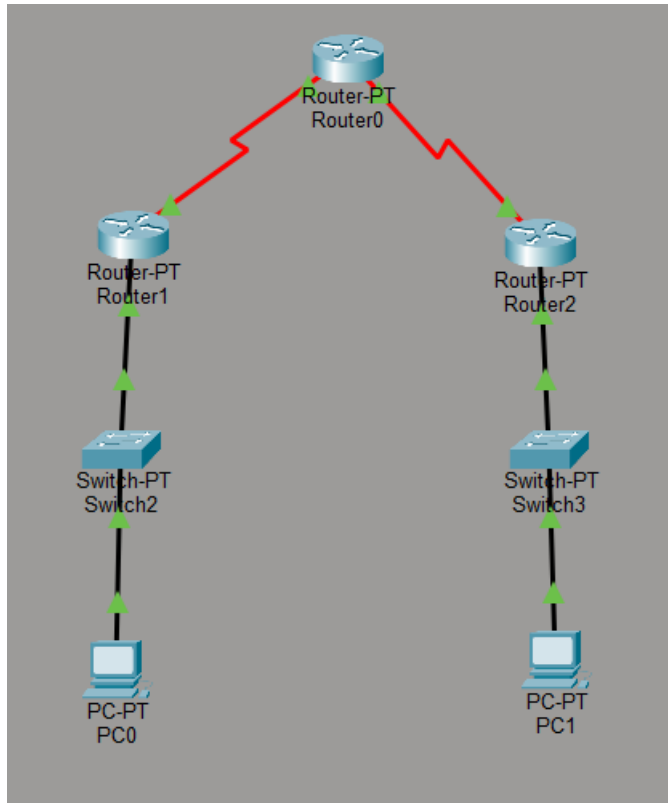
```
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
R   10.0.0.0/8 [120/1] via 20.1.1.1, 00:00:05, Serial3/0
C   20.0.0.0/8 is directly connected, Serial3/0
C   100.0.0.0/8 is directly connected, Serial2/0
R   172.16.0.0/16 [120/1] via 100.1.1.1, 00:00:03, Serial2/0
```

ASSIGNMENT 7

Routing using OSPF (Link State Routing)



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC0	PC1	ICMP		0.000	N	0	(edit)	
	Successful	PC0	PC1	ICMP		0.000	N	1	(edit)	

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	0	FULL/ -	00:00:34	100.1.1.1	Serial2/0
3.3.3.3	0	FULL/ -	00:00:32	20.1.1.1	Serial3/0

```
Router#show ip ospf database
```

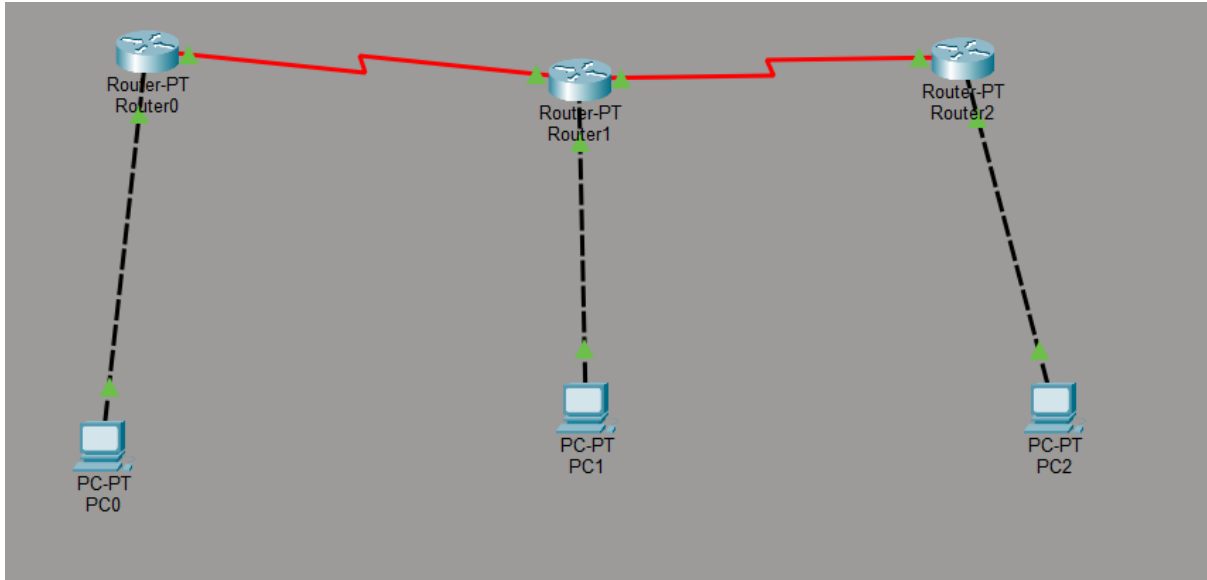
OSPF Router with ID (1.1.1.1) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	58	0x80000005	0x002ee0	4
2.2.2.2	2.2.2.2	58	0x80000004	0x003fd6	3
3.3.3.3	3.3.3.3	58	0x80000004	0x000f52	3

ASSIGNMENT 8

Configuration and Analysis of BGP Routing.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	15	(edit)	
	Failed	PC0	PC2	ICMP		0.000	N	16	(edit)	
	Successful	PC0	PC2	ICMP		0.000	N	17	(edit)	

```
Router#show ip bgp
BGP table version is 4, local router ID is 172.16.0.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*> 10.0.0.0/8      172.16.0.1           0      0      0 1 i
*> 40.0.0.0/8      172.14.0.2           0      0      0 79 i
*> 50.0.0.0/8      0.0.0.0              0      0 32768 i

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

B    10.0.0.0/8 [20/0] via 172.16.0.1, 00:00:00
B    40.0.0.0/8 [20/0] via 172.14.0.2, 00:00:00
C    50.0.0.0/8 is directly connected, FastEthernet0/0
C    172.14.0.0/16 is directly connected, Serial3/0
C    172.16.0.0/16 is directly connected, Serial2/0

Router#show ip bgp summary
BGP router identifier 172.16.0.2, local AS number 71
BGP table version is 4, main routing table version 6
3 network entries using 396 bytes of memory
3 path entries using 156 bytes of memory
2/2 BGP path/bestpath attribute entries using 368 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 1024 total bytes of memory
BGP activity 3/0 prefixes, 3/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent   TblVer   InQ OutQ Up/Down State/PfxRcd
172.16.0.1    4     1     12      12         4     0   0 00:09:21         4
172.14.0.2    4    79     13      12         4     0   0 00:10:13         4
```